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# Multi-fidelity graph neural networks for efficient power flow analysis under high-dimensional demand and renewable generation uncertainty

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ABSTRACT

The modernization of power systems faces uncertainties due to fluctuating renewable energy sources, electric vehicle expansion, and demand response initiatives. These uncertainties require consideration in power flow analyses by calculating power flow across a spectrum of generation and load values. Traditional power flow methods, which rely on iterative solutions of non-linear equation sets, become computationally burdensome under these uncertain conditions. Surrogate models have been used to reduces the computational cost of power flow analysis under uncertainty. Recently, graph neural networks (GNNs) have gained increasing attention as surrogates for power system simulations. However, GNNs, similar to other machine learning models, require significant amounts of training data. This paper proposes an innovative approach combining a multi-fidelity methodology with a GNN-based surrogate model for efficient power flow calculations. This model is trained using both high-fidelity power flow simulations and low-fidelity power flow simulations. Our multi-fidelity GNN model not only reduces the cost of generating training data, but also outperforms both single-fidelity GNN models trained solely on high-fidelity data and conventional neural network models. Additionally, the proposed model exhibits robustness to minor topology changes and achieves reasonable performance with unseen topologies. The model's effectiveness is validated through simulations on standard IEEE systems.

## 1. Introduction

Power flow analysis provides essential information regarding the flow of power through the branches and the voltage at the buses for a known set of generators and loads. This information renders power flow analysis a crucial tool for ensuring the security, reliability, and resilience of power systems [1].

Traditionally, power flow analysis involves solving a set of non-linear algebraic equations through iterative numerical methods [2,3]. However, the rapidly evolving energy landscape has new challenges that conventional power flow analysis struggles to address. Driven by the urgent need for decarbonization, there has been a significant shift in electricity generation; renewable energy sources such as solar and wind power generators are slowly replacing fossil fuel-based generators in the electric grids. These generators are inherently non-dispatchable, intermittently operating, and produce output with a high fluctuation because of the direct dependency on weather conditions [4–6]. Moreover, expanding electrification in various sectors such as electric vehicles, initiation of demand response programs, and disruptions due to cyber-attacks also add to the uncertainty and complexity of electric grids. Real-time decision-making in such a large, dynamic, complex, and stochastic system requires rapid and adaptive solutions, especially in critical situations, e.g., sudden grid topology alterations, unexpected demand surges, abrupt changes in generation, and cyber–physical attacks. Additionally, the uncertainty in generation and demand often necessitates running numerous power flow solutions to encompass the range of possible scenarios. Such probabilistic power flow analysis using traditional methods, particularly those based on the Newton–Raphson approach, can be challenging because of their computational cost, especially for large grids [7].

Addressing these emerging challenges requires innovative approaches that can adapt to the evolving nature of power systems. One promising avenue is the development of surrogate models that replace computationally expensive power flow model and provide fast probabilistic analysis [8,9]. In particular, artificial neural networks (NNs) have been widely used as surrogates in power flow analysis because of their ability in handling non-linearity [10,11]. However, conventional NNs often lack the ability to adapt to power grids with different sizes or topologies than those on which they were trained [12]. To overcome these shortcomings, recent research has been exploring the use of graph neural networks (GNNs) [13,14], by representing power grids as graphs, where the buses are the nodes (vertices) and the transmission lines connecting the buses are the edges of the graph [12,15–21].

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Since GNNs facilitate localized computations by concentrating on the immediate vicinity of individual nodes within the predominantly sparse graph structures of power systems, they are highly efficient. This approach involves aggregating information from the neighbors of a specific node, updating its representation based on this localized data. This method is particularly effective in power system graphs, where understanding the context – namely, the neighboring nodes and their interconnections – is essential to accurately interpret each node's attributes. This localized processing capability of GNNs ensures that even minor changes in the network topology, whether due to unforeseen line failures or planned rerouting activities, do not substantially impact the power flow calculations throughout the entire grid.

The application of AI-based methods, including GNNs, is often hindered by the significant computational cost associated with the requirement for large volumes of training data. High-fidelity training data, derived from detailed and accurate simulations like AC power flow methods, typically demands extensive computational resources. To address this challenge in the context of power flow analysis, we introduce an innovative approach using multi-fidelity data fusion in GNN models.

Our proposed multi-fidelity GNN (MF-GNN) framework integrates both low-fidelity and high-fidelity data to train the model efficiently. Low-fidelity data, obtained from the computationally cheaper DC power flow method, provides quick but less accurate estimations of phase angles. High-fidelity data, derived from the more detailed AC power flow method, includes complex voltage magnitudes and phase angles, offering higher accuracy but at a greater computational cost.

The proposed MF-GNN framework integrates a low-fidelity GNN (LF-GNN) and a high-fidelity GNN (HF-GNN). In this framework, the LF-GNN utilizes the low-fidelity data for initial approximations. The HF-GNN then refines these initial estimations using the high-fidelity data. Our model features a unique residual structure where the HF-GNN focuses on estimating the residuals (i.e., the differences between high-fidelity and low-fidelity estimations), a less complex task [22]. This design not only reduces computational demands but also enhances the model's performance, ensuring higher accuracy and robustness across scenarios with fixed, slightly altered, or entirely new grid topologies, thus improving generalizability.

We carry out three experiments to investigate the efficiency of our proposed MF-GNN model for power flow studies. First, we compare it to a single-fidelity GNN model that is only trained on data from high-fidelity simulations to explore the benefits of fusing data from multi-fidelity simulations. We also assess the GNN models' performance against two different NN setups to showcase the superiority of the GNN structure as a surrogate for power flow analysis. In our second experiment, we examine a scenario where the grid's topology is altered due to line removal. This experiment investigates the benefits of the proposed multi-fidelity GNN for contingency analysis. Lastly, we train the proposed model on various power systems, each with distinct topologies, to evaluate its ability to generalize effectively on unfamiliar grids.

The main contribution of the paper is summarized as follows:

- A novel multi-fidelity model for power flow calculation in the electric grid using GNN has been proposed. This model exploits the cheap computational cost of low-fidelity power flow data to reduce the required number of high-fidelity simulations to achieve accurate approximations.
- The experiments demonstrate that the multi-fidelity model is more effective than the single-fidelity model in adapting to unseen data. This is because utilizing multi-fidelity data enhances the information gained by the GNN model, leading to higher accuracy and stronger generalization performance.
- The proposed multi-fidelity model demonstrates enhanced robustness against small topology perturbations, such as a single line failure, compared to the single-fidelity model.

The rest of the paper is structured as follows. Background and related works concerning the application of graph neural networks in power flow analysis are presented in Sections 2 and 3, respectively. The proposed framework is outlined in Section 4. Experimental results are discussed in Section 5 and concluding remarks are provided in Section 6.

## 2. Background

### 2.1. Power flow analysis

The power flow (or load flow) [23,24] analysis in the electric power grid involves computation of the steady-state attributes associated with the buses (e.g., voltage phasor) and transmission lines (e.g., real and reactive power flow) using the grid topology, bus, branch, and transformer parameters, the active and reactive power output of the generators, and the real and reactive power demands corresponding to the loads connected to the grid. The power flow results are crucial for many applications related to grid operation, including scheduling of the generators, load dispatching, and maintenance-related issues such as load shedding and intentional islanding. The general form of power flow computation, also known as the *AC power flow (ACPF)*, which relates the real and reactive powers with the bus voltage phasors (magnitude and phase angles), is presented by the following set of non-linear equations:

$$P_i = |V_i| \sum_j |V_j|(G_{ij}\cos(\theta_i - \theta_j) + B_{ij}\sin(\theta_i - \theta_j)), \tag{1}$$

and

$$Q_i = |V_i| \sum_j |V_j|(G_{ij}\sin(\theta_i - \theta_j) - B_{ij}\cos(\theta_i - \theta_j)), \tag{2}$$

where $P_i$ and $Q_i$ denote the net real and reactive power injection at bus $i$, respectively, whereas $|V_i|$ and $\theta_i$ denote the magnitude and the phase angle of voltage at bus $i$, respectively. $G_{ij}$ and $B_{ij}$ are the conductance and susceptance of the transmission line connecting the buses $i$ and $j$, respectively.

Since the solutions of the above non-linear equations are computationally expensive for larger grids, a linearized version of the power flow analysis, known as *DC power flow (DCPF)*, is used to relate the active powers with the bus voltage phase angles assuming bus voltage magnitude of 1 per unit (p.u.) at all the buses and high reactance-resistance ratio. DCPF involves the computation of the following set of linear equations [25]:

$$P_i = \sum_j B_{ij}(\theta_i - \theta_j). \tag{3}$$

Although the results of DCPF often deviate from the realistic grid scenarios, due to its low computational cost, it is useful for a quick approximation of the grid attributes especially in applications where the active power is the main focus. In this paper, we propose a MF-GNN surrogate model for the power flow computation. This model offers a fast approximation of power flow, which facilitate probabilistic analysis in modern-day power systems characterized by high renewable energy and electric vehicle integration.

### 2.2. Graph neural networks

GNNs represent a significant advancement in deep learning for handling graph-structured data. These networks adeptly process not just node attributes but also the connections between nodes, capturing the full scope of the graph's structure [26,27]. This is particularly useful for analyzing graph signals, where the relationship between data points is as crucial as the data itself. GNNs also offer enhanced scalability compared to conventional neural networks, owing to their unique capability to directly capture and process complex interconnections in graphs. This improves the management of large datasets and their intricate relationships.

Central to the functioning of GNNs is the concept of message passing. This involves accumulating information from adjacent nodes to refine the representation of a specific node. The message passing procedure comprises two distinct steps: message aggregation and node update. Message aggregation specifically focuses on the gathering of information from neighboring nodes into a single message for a target node. The message aggregation is mathematically framed as [28]:

$$m_v^{(t)} = \sum_{u \in N(v)} M(h_u^{(t-1)}, h_v^{(t-1)}, e_{uv}). \tag{4}$$

In this context, $m_v^{(t)}$ symbolizes the aggregated message for node $v$ at layer $t$. $h_v^{(t)}$ designates the feature vector of node $v$ at layer $t$. $N(v)$ refers to nodes adjacent to $v$, and $M$ is a specific message function determining the exchange of messages between nodes based on their features and the edge features represented by $e_{uv}$.

Following the message aggregation phase, GNNs transition to the node update phase. This entails merging the gathered message with a node's current feature, culminating in an updated node feature. This integration is expressed as:

$$h_v^{(t)} = U(h_v^{(t-1)}, m_v^{(t)}). \tag{5}$$

Here, $U$ stands for the update function.

Some GNNs, drawing inspiration from the Weisfeiler–Lehman (WL) algorithm, have a distinct approach to updating node representations [29]. This involves a process where each node's representation is iteratively enhanced by gathering information from its neighboring nodes. In a WL-inspired GNN layer, this enhancement of node features is largely influenced by the local structure of the graph. For the power flow analysis in our study, we use the WL-GNN layer. The key advantage of this WL-GNN layer over general GNN layers is its ability to more effectively capture and integrate complex local graph structures, leading to potentially more accurate node representations.

The fundamental process within a WL-GNN layer, i.e., message passing, is expressed through the following equation:

$$a_u^{(t)} = \text{AGGREGATE}^{(t)}\left(\{h_v^{(t-1)} : v \in N(u)\}\right) \tag{6}$$

In this formula, $a_u^{(t)}$ denotes the aggregated feature for a particular node $u$ in the $t$th layer. The aggregation process typically involves summing, averaging, or finding the maximum of the features of the adjacent nodes, represented by $N(u)$.

Subsequently, each node's feature is updated in a consolidated manner as per the following equation:

$$h_u^{(t)} = \sigma\left(W^{(t)} \cdot \text{CONCAT}\left(h_u^{(t-1)}, a_u^{(t)}\right)\right) \tag{7}$$

Here, $h_u^{(t)}$ is the newly updated feature of node $u$ at the $t$th layer. $W^{(t)}$ stands for the trainable weight matrix assigned to that layer, while $\sigma$ is an activation function, such as ReLU. The CONCAT function serves to merge the existing feature of the node with the aggregated feature from the current layer.

## 3. Related works

### 3.1. Machine learning-based power flow modeling

In recent years, the field of power flow analysis has seen a significant shift towards the incorporation of machine learning techniques, notably through the application known as *solver learning* [10]. This approach utilizes simulation data to create mappings between power injections and bus voltages, effectively bypassing the traditional challenges of model rebuilding in PF analysis including high computational overhead, error accumulation, and non-convergence [30–32]. Inspired by the foundational principles of DC and AC power flows, several studies have tailored solver learning methods to simulate these equations, aiming to more accurately reflect the behavior of power systems.

Linear regression and support vector regression with polynomial kernels were among the initial algorithms employed. While linear regression [30,33] offers a basic model that was often considered simplistic, support vector regression, despite its ability to capture non-linear dependencies, is criticized for its high computational demand and scalability issues in larger networks [34]. As an alternative, neural networks like the multi-layer perceptron [35], radial basis function neural network [31,36], physics-informed neural network [10,37], and generative adversarial network [38] have been adopted for their ability to model complex, non-linear input–output relationships with greater accuracy and efficiency.

Recently, there has been a surge in methodologies utilizing GNNs for power grid computations. This trend is driven by the ability of GNNs to effectively learn and generalize complex relationships within power grids of varying sizes, thus offering enhanced scalability and adaptability beyond traditional training conditions.

Boltz et al. [18] proposed an GNN model for power flow balancing that uses a pre-computed correlation matrix to identify the neighboring nodes of grid buses. However, this approach has limitations as it does not account for the physical infrastructure of the power grid or the properties of power lines. Owerko et al. [17] introduced a GNN-based model designed to excel in solving a modified optimal power flow problem. This model has demonstrated effective results within its specific grid context. However, its architecture is closely linked to the characteristics of that particular grid, which limits its adaptability to handle topology changes within the same grid or to address different grid structures. Furthermore, their approach considered a simplified version of the optimal power flow problem which only computes active generator power outputs. Donon et al. [15] presented a GNN model that focuses on transmission lines as graph vertices rather than buses. This approach simplifies mathematical treatment by considering lines as the topological invariant. Their model, however, overlooked crucial branch attributes, pivotal for power flow calculations. In subsequent research, Donon et al. [16] developed an unsupervised model that placed particular emphasis on iterative voltage and phase angle approximation for each bus. This model was further enhanced by integrating a typed GNN [12] and attention mechanisms [39]. The integration of the typed GNN allowed the model to effectively handle different types of nodes and edges in the power system graph, enabling more context-aware analysis [12]. Additionally, the incorporation of attention mechanisms enabled the model to selectively focus on critical aspects of the data, improving its overall performance [39]. However, these studies employed GNNs for a task, i.e., linear approximation, that could potentially be just as efficiently handled by a simpler model such as NNs, given that essential calculations occur externally to the GNN [40]. Specifically, the GNN's role is to compute $\Delta\theta_i$ and $\Delta|V|_i$ from $\Delta P_i, \Delta Q_i$, a task akin to the operation conducted by the Newton–Raphson method with a linear approximation. For a comprehensive review of GNN methods for power flow calculation, readers are referred to [41].

### 3.2. Multi-fidelity modeling

The efficacy of any neural network model, including GNNs, heavily relies on the quality and quantity of training data. However, obtaining data from high-fidelity simulations for power flow analysis can be expensive and time-consuming. Some researchers have begun exploring the idea of multi-fidelity modeling, leveraging both low and high-fidelity data for improved results [42,43]. Yang et al. [43] introduced a multi-fidelity neural network (MF-NN) tailored for contingency analysis of power grids. The proposed network integrates low-fidelity data from DCPF approximations with high-fidelity data from ACPF data to enhance simulation accuracy. However, the effectiveness of their proposed multi-fidelity technique has not been justified by comparing it with a model that solely utilizes high-fidelity ACPF data. Moreover, due to the reliance of MFNN on a neural network structure, its generalizability to unseen grid topology is not guaranteed.

To address these challenges and expand upon the concept of multi-fidelity modeling, we introduce a multi-fidelity graph neural network (MF-GNN) tailored for power flow analysis. This innovative approach capitalizes on the inherent adaptability and scalability of GNNs, combining them with the computational benefits of multi-fidelity strategies. Unlike the MF-NN, our MF-GNN is designed with a keen focus on the structured nature of power systems, where the relationships between different components, such as buses, generators, and loads, can be effectively represented in a graph-based framework. This structural representation allows MF-GNN to naturally incorporate the topology of power grids as an integral part of the model, enhancing its generalizability and enabling its application to a broad range of grid configurations without the need for retraining. Furthermore, the proposed MF-GNN model distinguishes itself through a simplified yet efficient architecture, featuring fewer networks and trainable parameters compared to the MF-NN. Notably, MF-GNN incorporates only two GNNs versus the three neural networks required by the MF-NN [43]. This simplified architecture not only streamlines the model's complexity but also facilitates faster training and improved performance. The experimental results outlined in our study underscore the MF-GNN's superior accuracy and efficiency across various grid topologies, highlighting its potential as a robust tool for power flow analysis.

The subsequent sections will provide a detailed exposition of the MF-GNN's methodology, architecture, and comparative advantages, elucidating its contributions to the field of power systems analysis.

## 4. Methodology

This section offers a thorough exploration of the proposed MF-GNN methodology for computing power flow within electrical grids. To utilize GNN techniques with power system data, it is essential to express the measurements as graph data or graph signals [44,45]. We first discuss how the power grid and corresponding data measurements can be conceptualized within a graph-structured data framework, treating buses as nodes and branches as edges. Next, we delve into the design of the MF-GNN model for power flow calculations. This includes explanation on the model's architecture, the formulation of the loss function, and the training process.

### 4.1. Graph representation of power system

In this study, we model an electric power grid comprising $N$ buses and $L$ transmission lines as a graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$. The set $\mathcal{V}$ represents the nodes of the graph, corresponding to the buses, and the set $\mathcal{E}$ comprises edges that represent the transmission lines connecting the buses. An edge $e_{ij}$ is an element of $\mathcal{E}$ if there is a transmission line connecting bus $i$ to bus $j$. We denote the number of nodes and edges by $|\mathcal{V}| = N$ and $|\mathcal{E}| = L$, respectively, where $|\cdot|$ indicates the cardinality of a set. The adjacency matrix for the graph $\mathcal{G}$ is defined as $\mathbf{A} = [a_{ij}]$, with $a_{ij} = 1$ if there is an edge $e_{ij} \in \mathcal{E}$, and $a_{ij} = 0$ otherwise.

Each node $v_i \in \mathcal{V}$ is characterized by a feature vector $\mathbf{x}_i$ that encapsulates the attributes of bus $i$. These attributes are defined as:

$$\mathbf{x}_i = \left[P_i^d, Q_i^d, P_i^g, V_i^s\right], \tag{8}$$

where $P_i^d$, $Q_i^d$, $P_i^g$, and $V_i^s$ respectively denote the active power demand, reactive power demand, generated active power, and voltage set-point at bus $i$. It is important to recognize the differences among buses based on their functions. For example, load buses typically have their generation-related elements ($P_i^g$ and $V_i^s$) set to zero. In this study, to reflect the integration of renewable energy sources, $P_i^g$ is assigned a nonzero value for all load buses. Additionally, for generation buses, the demand elements, $P_i^d$ and $Q_i^d$, are set to zero. Notably, low-fidelity DC simulation considers reactive power demands to be zero at all buses. Our proposed method utilizes this node feature information within a GNN framework to estimate the bus voltage magnitude, $|V|$, at load buses, and phase angle, $\theta$, at all buses.

### 4.2. Multi-Fidelity Graph Neural Network (MF-GNN)

In this study, we propose a novel residual GNN architecture tailored for multi-fidelity power flow modeling, as showcased in Fig. 1. The proposed MF-GNN model is characterized by two primary components: LF-GNN and HF-GNN. The purpose of these components is to effectively utilize both low-fidelity data obtained by DCPF method and high-fidelity data obtained by ACPF method. The proposed MF-GNN can replace traditional power flow model for rapid computation of bus voltage phasors.

The low-fidelity component of MF-GNN is trained using only DCPF approximations, which considers the voltage magnitudes for all buses to be 1 p.u. Consequently, the LF-GNN is designed to only provide low-fidelity approximations of the phase angle. The LF-GNN operates as:

$$\begin{aligned} H_{\mathrm{LF}}^{(0)} &= \mathbf{x}, \\ H_{\mathrm{LF}}^{(l+1)} &= \mathrm{ReLU}(\mathrm{GNN}^{(l)}(H_{\mathrm{LF}}^{(l)}, \mathbf{A})), \\ \hat{\boldsymbol{\Theta}}_{\mathrm{LF}} &= \mathrm{GNN}^{(L-1)}(H_{\mathrm{LF}}^{(L-1)}, \mathbf{A}), \end{aligned} \tag{9}$$

where $H_{\mathrm{LF}}^{(l)}$ represents the node features at layer $l$ of LF-GNN, $\mathbf{A}$ is the adjacency matrix representing the power system graph, $\mathrm{GNN}^{(l)}$ denotes the $l$th GNN layer, $\hat{\boldsymbol{\Theta}}_{\mathrm{LF}}$ is the estimated phase angles.

The loss function in the LF-GNN model is based on the mean squared error (MSE). This function calculates the average of the squared differences between the approximated phase angles ($\hat{\boldsymbol{\Theta}}_{\mathrm{LF}}$) and the actual phase angles calculated using DCPF ($\boldsymbol{\Theta}_{\mathrm{LF}}$), averaged over both the number of buses $N$ and the number of low-fidelity training data samples $T_l$. That is:

$$\mathcal{L}_{\mathrm{LF}} = \frac{1}{T_l N} \sum_{i=1}^{T_l} \sum_{j=1}^{N} (\hat{\boldsymbol{\Theta}}_{\mathrm{LF},j}^{(i)} - \boldsymbol{\Theta}_{\mathrm{LF},j}^{(i)})^2, \tag{10}$$

where, $\hat{\boldsymbol{\Theta}}_{\mathrm{LF},j}^{(i)}$ and $\boldsymbol{\Theta}_{\mathrm{LF},j}^{(i)}$ represent the estimated and actual phase angles at bus $j$ for the $i$th training sample, respectively.

The HF-GNN is designed to enhance the estimations of the LF-GNN. It takes the phase angles estimated by the LF-GNN and pairs them with a vector of ones to represent voltage magnitudes, denoted by $\mathbf{1}$ in Fig. 1. This combined data, along with the original node features, is then used as input for the HF-GNN. This model passes the input through multiple GNN layers to evaluate a high-fidelity version of voltage magnitudes and phase angles. The HF-GNN employs a residual design, meaning it estimates only the differences between high-fidelity and low-fidelity estimations, represented as $\mathbf{R}_\theta$ and $\mathbf{R}_V$. The final refined estimations are obtained by adding these differences to the initial low-fidelity estimations. Since these differences are typically small, the network can remain relatively simple, yet still minimize approximation errors. HF-GNN can be represented as:

$$\begin{aligned} H_{\mathrm{HF}}^{(0)} &= \mathrm{Concat}(\mathbf{x}, \mathbf{1}, \hat{\boldsymbol{\Theta}}_{\mathrm{LF}}), \\ H_{\mathrm{HF}}^{(l+1)} &= \mathrm{ReLU}(\mathrm{GNN}^{(l)}(H_{\mathrm{LF}}^{(l)}, \mathbf{A})), \\ \mathbf{R}_V, \mathbf{R}_\theta &= \mathrm{GNN}^{(L-1)}(H_{\mathrm{HF}}^{(L-1)}, \mathbf{A}), \\ \hat{\mathbf{V}}_{\mathrm{HF}} &= \mathbf{R}_V + \mathbf{1} \\ \hat{\boldsymbol{\Theta}}_{\mathrm{HF}} &= \mathbf{R}_\theta + \hat{\boldsymbol{\Theta}}_{\mathrm{LF}}. \end{aligned} \tag{11}$$

The HF-GNN model employs a loss function that precisely estimates voltage magnitudes and phase angles, using a MSE approach. This function calculates the average of the squared differences between the estimated voltage magnitudes ($\hat{\mathbf{V}}_{\mathrm{HF}}$) and phase angles ($\hat{\boldsymbol{\Theta}}_{\mathrm{HF}}$) against their actual values obtained from ACPF ($\mathbf{V}_{\mathrm{HF}}$ and $\boldsymbol{\Theta}_{\mathrm{HF}}$), averaged over both the number of buses $N$ and the number of high-fidelity training simulations $T_h$. That is:

$$\mathcal{L}_{\mathrm{HF}} = \frac{1}{T_h N} \sum_{i=1}^{T_h} \sum_{j=1}^{N} \left[(\hat{\mathbf{V}}_{\mathrm{HF},j}^{(i)} - \mathbf{V}_{\mathrm{HF},j}^{(i)})^2 + (\hat{\boldsymbol{\Theta}}_{\mathrm{HF},j}^{(i)} - \boldsymbol{\Theta}_{\mathrm{HF},j}^{(i)})^2.\right] \tag{12}$$

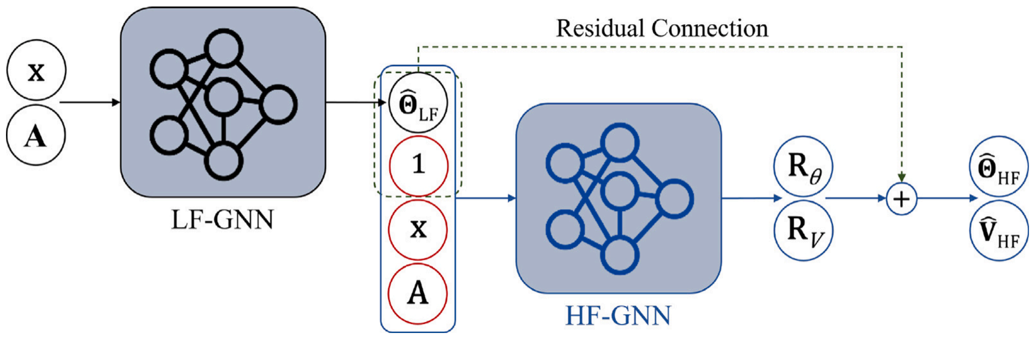



**Fig. 1.** Architecture of the proposed MF-GNN model, illustrating both the LF-GNN and HF-GNN components. The LF-GNN, using node features $\mathbf{x}$ and adjacency matrix $\mathbf{A}$, estimates low-fidelity phase angles, $\hat{\mathbf{\Theta}}_{\mathrm{LF}}$. The HF-GNN refines these estimations, producing high-fidelity estimates for voltage magnitudes, $\hat{\mathbf{V}}_{\mathrm{HF}}$, and phase angles, $\hat{\mathbf{\Theta}}_{\mathrm{HF}}$, by estimating residuals $\mathbf{R}_\theta$ and $\mathbf{R}_V$, which are added to the initial LF-GNN outputs.

Here, $\hat{\mathbf{V}}^{(i)}_{\mathrm{HF},j}$ and $\hat{\mathbf{\Theta}}^{(i)}_{\mathrm{HF},j}$ denote the estimated voltage magnitude and phase angle at bus $j$ for the $i$th training sample, respectively, and $\mathbf{V}^{(i)}_{\mathrm{HF},j}$ and $\mathbf{\Theta}^{(i)}_{\mathrm{HF},j}$ are their actual counterparts obtained from ACPF.

It must be highlighted that for cases where the computational budget allows having DC simulation, the proposed MF-GNN framework can be adopted to replace the LF-GNN component with an exact DCPF solver. Specifically, the genuine DC approximation is utilized as the input for the HF-GNN component, instead of approximations from LF-GNN. Consequently, this modification entails substituting $\mathbf{\Theta}_{\mathrm{LF}}$ for $\hat{\mathbf{\Theta}}_{\mathrm{LF}}$ in Eq. (11). This adjustment can significantly reduce the error accumulation and propagation inherent within the two GNN components, thereby enhancing overall accuracy.

## 5. Results and discussion

In this section, we describe the experimental design used to validate the proposed MF-GNN method for power flow analysis and present the results we obtained.

### 5.1. Experiment setup

We conduct three experiments to evaluate the performance of our proposed MF-GNN model in various scenarios. The first experiment is a straightforward test of the MF-GNN model's ability to calculate power flow. We focus on how accurately it estimates bus voltage magnitudes and phase angles, assuming a fixed grid topology. The second experiment aims to determine the model's robustness against minor topology disruptions, specifically those caused by single-line failures. In the third experiment, we test the model's adaptability to training with varying grid topologies. This involves altering the grid's adjacency matrix, $\mathbf{A}$, to include a range of IEEE standard grid topologies.

#### 5.1.1. Baseline models

Across all three experiments, we compare the MF-GNN model's performance with that of a single-fidelity GNN (SF-GNN) model, which is trained only with high-fidelity data. Additionally, we use two fully connected neural network-based methods, a local NN and a global NN, as benchmarks for our comparisons. The local NN focuses on estimating the target variables, i.e., voltage magnitude and phase angle, at each individual bus. Specifically, this model utilizes five specific features corresponding to that bus, while disregarding the topology and characteristics of the rest of the grid. In contrast, the global NN estimates the voltage magnitudes and phase angles for all buses in the grid simultaneously, incorporating attributes from every bus. This is achieved by merging the attributes of all buses into a single, one-dimensional array. This method allows the model to implicitly consider the entire grid's context, although it overlooks the interconnections between the buses. Furthermore, our comparison extends to include MF-NN proposed by [43], adding another layer of depth to our benchmarking by assessing our model against an existing multi-fidelity approach.

#### 5.1.2. Generation of data

Both the data for training the model and testing the model's performance by the three experiments mentioned above are generated using `PyPower` [46] on the standard IEEE grids (see Table 1 for detailed description). The non-convergence cases of the ACPF or DCPF are excluded from consideration for generating data samples.

For data generation, we execute power flow simulations (ACPF for high-fidelity and DCPF for low-fidelity data) in `PyPower`, treating both active and reactive power demands at all buses as random variables drawn from Gaussian distributions. The means of these distributions are equal to the original load values, with standard deviations set at 5% of these mean values. Additionally, active power generation at each load-only bus is also modeled as a random variable with a Gaussian distribution to simulate the integration of distributed renewable power generation. Here, the mean is set at 10% of the original active power demand value of that bus, with the standard deviation defined as 5% of this mean generation value. Given the large number of buses in the power system, this results in a high-dimensional space of demand and generation inputs.

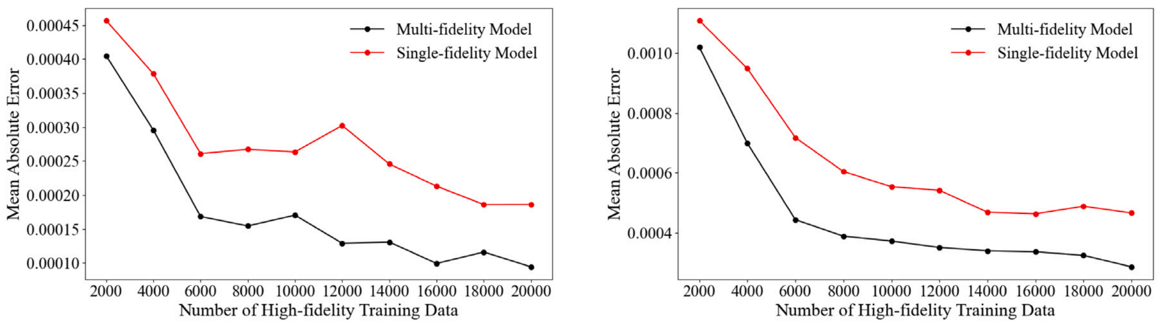
For a fair comparison between SF-GNN and MF-GNN models, we align the computational costs of generating their training data. Since ACPF analysis is approximately five times more computationally intensive than DCPF analysis for the considered test grids, the SF-GNN model uses more high-fidelity data than the MF-GNN model to balance their computational costs.

#### 5.1.3. Training process

All our deep learning models are implemented in PyTorch [47], and we employ the PyTorch Geometric library [48] to construct the GNN architectures. To conduct the hyperparameter search, we utilized `Optuna`, a Python package for hyperparameter optimization [49], which provides a flexible framework for hyperparameter tuning using techniques like Bayesian optimization [50] and Tree-structured Parzen Estimators [51], integrating seamlessly with PyTorch. We split the data into 70% for training, 20% for validation, and 10% for testing, and selected hyperparameters that yielded low validation errors.

In all the following experiments, the structure of the hidden layers is kept consistent across both the LF-GNN and HF-GNN elements within the MF-GNN model, as well as in the SF-GNN model. However, it is important to note that there are distinct input and output layers for each model, as illustrated in Fig. 1. This structure comprises three WL-GNN layers [29], each consisting of 64 hidden units and employing sum aggregation. ReLU activation functions are employed across these layers, except in the final layer. The models undergo training for 1000 epochs, leveraging the Adam optimization algorithm [52] with a learning rate of 0.001. An NVIDIA RTX 4090 GPU supports the training process.

The modular design of our proposed MF-GNN framework allows for the flexibility to incorporate branch attributes by replacing the WL-GNN layer with other GNN layers that support edge features, such




(a) Voltage Magnitude

(b) Voltage Phase Angle

**Fig. 2.** Comparison of mean absolute errors in voltage estimation for the IEEE 14-Bus system, using multi-fidelity (MF-GNN) and single-fidelity (SF-GNN) models. The $x$-axis indicates the number of high-fidelity simulations used to train the MF-GNN model, which additionally leverages twice as many low-fidelity simulations. Given that the computational cost of high-fidelity simulations is approximately five times that of low-fidelity simulations, the SF-GNN model, which exclusively uses high-fidelity simulations, is allocated 40% more high-fidelity simulations than the MF-GNN at each comparison point.

**Table 1**
Overview of IEEE test systems [54].

| Test system | No. of Buses | No. of generators | No. of branches |
|---|---|---|---|
| Case 9 | 9 | 3 | 9 |
| Case 14 | 14 | 5 | 20 |
| Case 24 | 24 | 33 | 38 |
| Case 30 | 30 | 6 | 41 |
| Case 39 | 39 | 10 | 34 |
| Case 118 | 118 | 54 | 186 |
| Case 2383 | 2383 | 327 | 2896 |

as the graph isomorphism network layer [53]. However, our initial experiments showed that including branch attributes as edge features did not significantly enhance the model's performance, so we opted to maintain the framework's efficiency and simplicity by excluding them in the final model.

### *5.1.4. Evaluation metrics*

We evaluate the performance of the proposed mode (MF-GNN) and the benchmark models in each experiment on the test data set using mean absolute error (MAE) defined as:

$$\text{MAE} = \frac{1}{n}\sum_{i=1}^{n} \left|\hat{y}_i - y_i,\right| \tag{13}$$

where, $n$ represents the total number of data points in the test set, $\hat{y}_i$ and $y_i$ are the estimated and actual values for the $i$th data point (comprising both the bus voltage magnitudes and phase angles), respectively.

### *5.2. Performance evaluation on fixed grid topology*

In this initial experiment, the models undergo training exclusively using instances featuring identical grid topology. These instances are generated through the data generation procedure detailed in Section 5.1.2. This represents the most straightforward scenario, where the models are solely required to estimate the system state for different values of loads on the buses, under a fixed system topology.

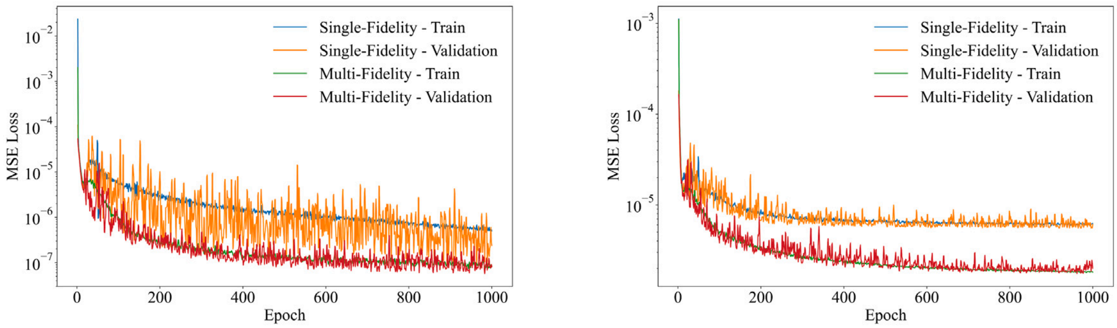
Fig. 2 showcases the accuracy of the proposed MF-GNN in voltage magnitude and phase angle estimations across varying sizes of training data, ranging from 2000 to 20,000 samples. It can be seen that MF-GNN demonstrates superior performance over SF-GNN in estimating both phase angles and voltage magnitudes. This enhancement is primarily due to the integration of information from the low-fidelity model, as well as the residual architecture of MF-GNN. Notably, the enhancement in phase angle estimation is more significant, as depicted in Fig. 2(b). Specifically, the LF-GNN uses a DCPF approximation that concentrates exclusively on phase angle estimation, setting the voltage magnitude to a uniform 1 p.u. across all buses. As a result, the estimations from the LF-GNN, when used as supplementary input for the MF-GNN, primarily improve its phase angle estimation accuracy. From this analysis, we have chosen to utilize 6000 high-fidelity training samples for further experiments.

Moreover, this figure illustrates that the MF-GNN model, trained using 6000 HF and 12,000 LF data points, surpasses the accuracy that a SF-GNN model can achieve, even with 28,000 HF data points. Furthermore, the cost-effectiveness of the MF-GNN is highlighted by the substantial savings in computational time: it takes only 38.2 s to generate training data for the MF-GNN with combined HF and LF data, compared to 127.4 s for the SF-GNN with 28,000 HF data points, representing a 70% reduction in computational costs. This efficiency gain is even more pronounced in larger systems, such as Polish 2383-Bus System, where generating 6000 HF and 12,000 LF data for training the MF-GNN requires only 61 s, whereas producing 28,000 HF data points for the SF-GNN takes 1470 s.

The learning trajectories of the MF-GNN and SF-GNN models throughout 1000 epochs is illustrated in Fig. 3. Fig. 3(a), shows the differences in training and validation losses of these models when estimating voltage magnitude. Notably, the MF-GNN model consistently records lower losses for both training and validation relative to the SF-GNN model. A similar trend is evident for phase angle estimations in Fig. 3(b). The MF-GNN model registers notably reduced losses within the same epoch span in comparison to the SF-GNN model.

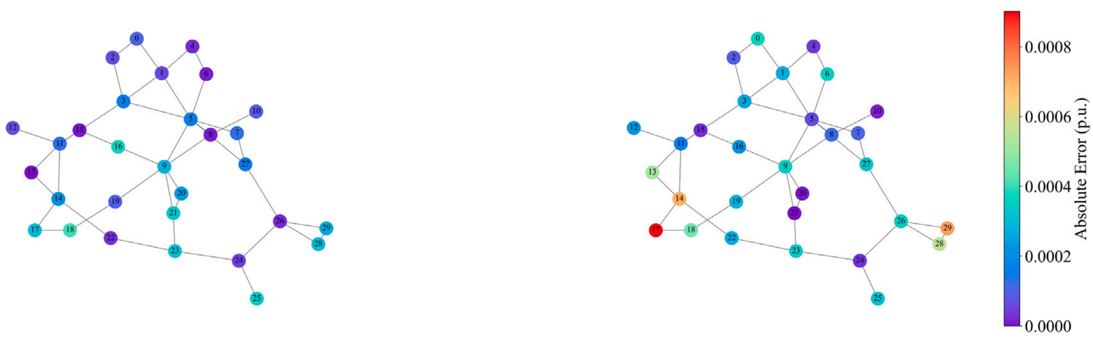
The absolute errors of voltage magnitude estimation for a single sample from the IEEE 30-Bus system using MF-GNN and SF-GNN have been illustrated in Fig. 4. It can be observed that the MF-GNN exhibits consistently lower absolute errors throughout the network in comparison to the SF-GNN. Similarly, the absolute errors in voltage phase angle estimation for the same sample from the IEEE 30-Bus system is demonstrated in Fig. 5, where a side-by-side comparison between the MF-GNN and SF-GNN, reveals that MF-GNN consistently outperforms SF-GNN in terms of approximation error.

Fig. 6 shows the estimated probability distribution function (PDF) of voltage magnitude and phase angle using MF-GNN and SF-GNN models for Bus 15 in the IEEE 30-Bus system under uncertain load and generation at different buses. Specifically, to accurately simulate the integration of uncertain loads and distributed generators, we model the active and reactive loads at each bus as random variables following Gaussian distributions, where the means are set to the original respective values and the standard deviations are 5% of these means. Additionally, active power generation from distributed renewable generations at each load bus is also modeled as a random variable with a Gaussian distribution. The mean for generation is set at 10% of the original active power demand value at that bus, with a standard deviation of 5% of this mean generation value. It can be seen that



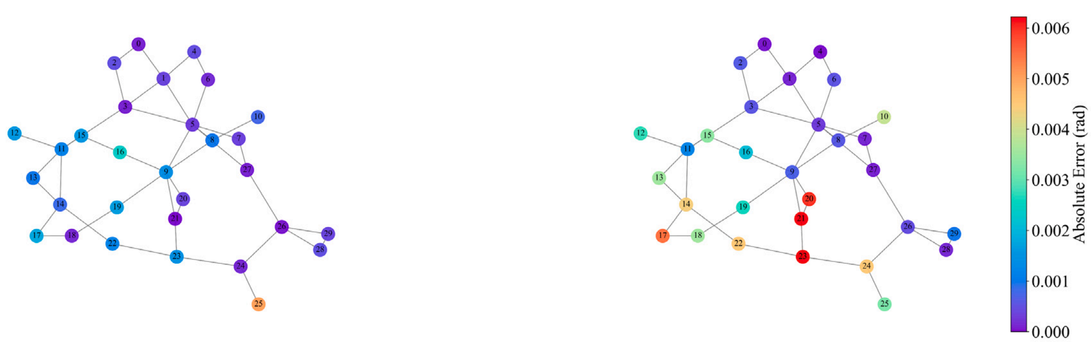

(a) Voltage Magnitude    (b) Voltage Phase Angle

**Fig. 3.** Comparison of single-fidelity and multi-fidelity models training and validation MSE losses against epochs for voltage estimation for the IEEE 30-Bus system.




(a) Multi-Fidelity Model    (b) Single-Fidelity Model

**Fig. 4.** Absolute errors of voltage magnitude approximation for a single sample from IEEE 30-Bus system.

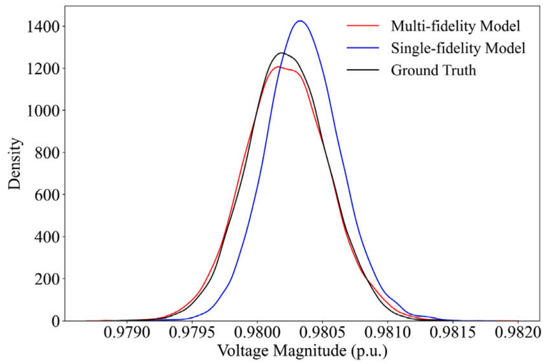



(a) Multi-Fidelity Model    (b) Single-Fidelity Model

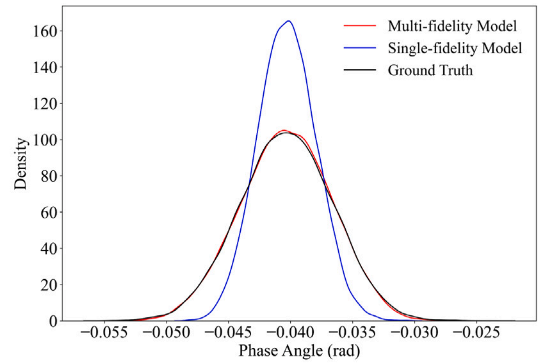
**Fig. 5.** Absolute errors of voltage phase angle approximation for a single sample from IEEE 30-Bus system.

the approximated distributions using MF-GNN align more closely with the ground truth. This makes MF-GNN a more reliable tool in many applications related to the monitoring and security of power systems, e.g., predicting and identifying voltage instability in the grid, detecting different types of cyber-attacks on power system measurements, detecting and locating line failure or other physical events in the grid.

The proposed model is further compared with two types of fully connected deep neural networks (DNNs): *local DNN* and *global DNN*, as well as the MF-NN model proposed by [43], and the support vector regression (SVR) model, which employs a non-linear radial basis function (RBF) kernel. Tables 2 and 3 show the accuracy of different types of models across multiple IEEE case systems. We observe that the proposed MF-GNNs consistently yield the lowest MAE for both voltage magnitude and phase angle across all test cases. This validates earlier observations and discussions, highlighting the excellence of MF-GNNs, particularly attributed to its unique architectural design and its ability to exploit grid structure as well as computationally cheap low-fidelity simulations. The results for MF-GNN with exact DCPF inputs also show that in cases where computational constraints allow DCPF simulation, replacing the LF-GNN component with a DCPF solver significantly decreases the MAE of the proposed MF-GNN. The SF-GNN follows closely, with MAEs generally larger than the MF-GNN but still significantly lower than the neural network models. This indicates the merits of graph-structured data and the capability of GNNs to capture relationships among the bus attributes in the grid.




(a) Voltage Magnitude




(b) Voltage Phase Angle

**Fig. 6.** Estimated PDFs voltage for Bus 15 in the IEEE 30-Bus system.

**Table 2**
Comparison of MAE for voltage magnitude across different models. All values are in per unit and scaled by $\times 10^{-3}$.

| Case | MF-GNN (w/ DCPF inp.) | MF-GNN | SF-GNN | Local DNN | Global DNN | MF-NN (w/ DCPF inp.) | MF-NN | SVR |
|---|---|---|---|---|---|---|---|---|
| Case 9 | 0.157 | 0.205 | 0.217 | 84.580 | 34.450 | 3.527 | 4.702 | 36.018 |
| Case 14 | 0.144 | 0.155 | 0.267 | 83.282 | 32.891 | 5.348 | 5.360 | 5.124 |
| Case 30 | 0.149 | 0.234 | 0.390 | 89.235 | 46.509 | 5.715 | 3.723 | 38.857 |
| Case 118 | 0.312 | 0.687 | 1.194 | 84.036 | 20.977 | 2.177 | 2.494 | 12.855 |
| Case 2383 | 2.827 | 6.470 | 6.882 | 78.634 | 64.112 | 4.232 | 24.585 | 8.632 |

**Table 3**
Comparison of MAE for phase angle across different models. All values are in radians and scaled by $\times 10^{-3}$.

| Case | MF-GNN (w/ DCPF inp.) | MF-GNN | SF-GNN | Local DNN | Global DNN | MF-NN (w/ DCPF inp.) | MF-NN | SVR |
|---|---|---|---|---|---|---|---|---|
| Case 9 | 0.058 | 1.587 | 3.432 | 61.549 | 9.449 | 1.251 | 8.756 | 75.208 |
| Case 14 | 0.075 | 0.389 | 0.604 | 66.266 | 5.664 | 1.172 | 3.403 | 39.607 |
| Case 30 | 0.094 | 1.004 | 1.669 | 16.105 | 3.573 | 0.751 | 2.944 | 69.175 |
| Case 118 | 0.392 | 6.379 | 6.940 | 109.698 | 12.212 | 3.618 | 10.376 | 257.406 |
| Case 2383 | 2.212 | 78.589 | 93.502 | 235.524 | 24.290 | 22.832 | 42.762 | 221.378 |

**Table 4**
Comparison of number of trainable parameters and inference times for multi-fidelity models in the Polish 2383-Bus System.

| Model | MF-GNN (w/ DCPF inp.) | MF-GNN | MF-NN (w/ DCPF inp.) | MF-NN |
|---|---|---|---|---|
| Number of trainable parameters | 35,330 | 70,148 | 159,012,827 | 159,084,321 |
| Inference time [s] | 0.0049442 | 0.0002338 | 0.0264266 | 0.0222719 |

On the other hand, the MF-NN model, while not performing as well as MF-GNN, shows notable improvement when DCPF inputs are integrated, underscoring the general effectiveness of incorporating exact low-fidelity inputs in a multi-fidelity setting to minimize potential inherent error propagation in such settings. This observation points to the broader applicability of DCPF integration in enhancing model performance across different architectures. Following this, we include a comparison of inference times and the number of trainable parameters in Table 4, particularly highlighting the differences between models with and without the DCPF input. The integration of the DCPF solver, expectedly increases inference time. However, the increment in inference time is rather small and most likely acceptable, depending on the application, considering the significant enhancements in model accuracy.

The local DNN, which makes estimations for each bus based solely on its specific input features, unsurprisingly records the highest errors, emphasizing the importance of a global context when estimating individual bus outputs. The global DNN, while encompassing information from all buses, is still unable to match the performance of GNNs. This is attributed to the absence of explicit topological and connectivity information about the grid components that the GNN inherently captures. Consequently, in the next experiments that involve changing topologies, DNNs will not be included for comparison.

### 5.3. Robustness evaluation against small topology perturbation

In this experiment, the models undergo both training and testing using variations of a baseline grid. These variations are created by implementing the data generation procedure outlined in Section 5.1.2, along with disconnections created by intentional tripping of a single transmission line in the network. This configuration is designed to simulate situations where the grid experiences disruptions due to line outages or maintenance activities. It is crucial for the GNN-based power flow solver to be robust against such scenarios and provide precise estimations in applications such as contingency analysis.

Table 5 showcases a comparison between our proposed MF-GNN models and the SF-GNN model across different scenarios of topological disturbances in power grids. Notably, MF-GNNs consistently outperform SF-GNN in estimating voltage magnitude and phase angle across all tested cases. It must be noted that WSCC 9-Bus System, which is a radial network, is excluded from this analysis. This is because radial networks have single paths between points, making them highly sensitive to line failures, often leading to complete loss of connectivity. For all other cases, the outcomes indicate that MF-GNNs, particularly with DCPF inputs enhancing the HF-GNN component, demonstrate reduced MAE and increased reliability and accuracy compared to SF-GNN. These advantages highlight the potential of MF-GNNs as more effective tools for contingency analysis in power systems during grid disturbances.

**Table 5**
Comparison of MAE for MF-GNNs, and SF-GNN on test set with topological perturbations. All values, with magnitudes in per unit and phase angles in radians, are scaled by $\times 10^{-3}$.

| Case | MF-GNN w/ DCPF inp. | | MF-GNN | | SF-GNN | |
|---|---|---|---|---|---|---|
| | $V$ | $\theta$ | $V$ | $\theta$ | $V$ | $\theta$ |
| Case 14 | 0.424 | 0.337 | 1.099 | 3.473 | 1.490 | 4.629 |
| Case 30 | 0.932 | 0.448 | 1.481 | 3.346 | 1.627 | 4.579 |
| Case 118 | 0.506 | 1.441 | 0.910 | 9.559 | 1.255 | 10.023 |
| Case 2383 | 2.773 | 2.130 | 5.795 | 69.910 | 6.142 | 87.728 |

**Table 6**
Comparison of MAE for MF-GNNs, and SF-GNN on test set across training and unseen grid topologies. All values, with magnitudes in per unit and phase angles in radians, are scaled by $\times 10^{-3}$.

| Case | MF-GNN w/ DCPF inp. | | MF-GNN | | SF-GNN | |
|---|---|---|---|---|---|---|
| | $V$ | $\theta$ | $V$ | $\theta$ | $V$ | $\theta$ |
| Case 9 | 0.186 | 0.118 | 0.386 | 3.602 | 0.810 | 3.880 |
| Case 14 | 0.252 | 0.170 | 0.681 | 1.306 | 0.916 | 2.008 |
| Case 30 | 0.185 | 0.206 | 0.395 | 1.565 | 0.598 | 2.413 |
| Case 118 | 0.124 | 0.342 | 0.638 | 6.270 | 0.847 | 6.788 |
| Case 24 | 50.312 | 42.820 | 66.910 | 380.465 | 281.851 | 878.083 |
| Case 39 | 63.744 | 49.400 | 95.287 | 399.049 | 93.749 | 475.755 |

### *5.4. Performance evaluation across multiple grid topologies*

In this experiment, we train GNN models concurrently on datasets derived from power flows across four distinct grids: WSCC 9-Bus System, IEEE 14-Bus System, IEEE 30-Bus System, and IEEE 118-Bus System. A key feature of GNNs is their ability to integrate the adjacency matrix of each grid, which captures the unique topology of each grid, directly into the model's input. This enables GNNs to handle and learn from datasets with different structural characteristics simultaneously. Consequently, despite the varied topologies of these systems, the models can be trained concurrently. The test dataset encompasses data from these four systems, with additional data generated from IEEE 24-Bus System and IEEE 39-Bus System – two grids not included in the training set – to evaluate the model's generalization capabilities to new, unseen grid structures.

Table 6 presents MAEs for MF-GNN and SF-GNN models across different grid cases. The results indicate that MF-GNNs consistently outperform SF-GNN, not only for cases included in the training data (i.e., 9-, 14-, 30-, and 118-Bus systems) but also for the 24-bus and 39-bus systems, which represent a completely unseen scenario. MF-GNN demonstrates superior performance in these scenarios primarily because it leverages a more extensive set of computationally inexpensive training data compared to SF-GNN. Additionally, the HF-GNN component of MF-GNN, which employs a simple structure, is inherently less prone to overfitting, contributing to the robustness of MF-GNN when encountering new topologies. This enhanced generalization ability is further observed in power flow analysis for unseen scenarios, especially when data from DCPF is available, substantiating MF-GNN's advantage in handling diverse and previously unseen grid topologies.

## 6. Conclusion

Surrogate models are increasingly replacing traditional numerical methods for power flow computation to address the challenges posed by the size, complexity, and uncertainties inherent in modern power systems. In this paper, we propose a multi-fidelity GNN-based technique for power flow calculation in electric power grids. By explicitly considering the topology and interconnection among the components of the grid through the graph, this method outperforms other fully connected neural network-based power flow techniques in terms of voltage phasor estimation accuracy. Moreover, the multi-fidelity approach tackles the computational cost of required training data generation by efficiently infusing DCPF-based low-fidelity data into ACPF-based high-fidelity data. Due to the residual-based training mechanism, the MF-GNN achieves even better accuracy than GNN models trained only with high-fidelity data. Our experiments also reveal that the proposed MF-GNN is capable of estimating the voltage phasor with reasonable accuracy under small perturbations in the grid topology and even for new grid structures that are not present in the training data.

While the proposed multi-fidelity GNN model demonstrates significant improvements in efficiency and accuracy, it is not without limitations. One primary limitation is the long-range communication problem inherent in GNNs. This issue arises because GNNs tend to have difficulty capturing dependencies between distant nodes in large graphs, which can affect the accuracy of power flow calculations in extensive grid networks. To mitigate this limitation, edge augmentation techniques will be integrated into the model, enhancing the ability to capture long-range interactions between nodes. Another limitation is the use of basic models for renewable energy resources and fluctuating load demands. More realistic models that consider weather and related uncertainties would enhance the dataset's realism.

The proposed multi-fidelity power flow calculation technique has been proven accurate, efficient, and robust through extensive simulations on IEEE standard systems. However, implementing this technique in real-world scenarios poses challenges. Real-world generators, loads, and inverter-based resources may exhibit more sophisticated behavior, requiring advanced modeling. This necessitates subtle tuning of model parameters during data generation. Additionally, while the model is robust to line perturbations, knowing the exact failure location remains challenging. Integrating MF-GNN surrogates into existing workflows also involves ensuring compatibility with current software, hardware, and regulatory requirements.

Future research can explore transfer learning or curriculum learning [55] to guide the model training process and integrate optimal sampling techniques [56] to reduce the number of required high-fidelity simulations. These approaches could help address some of the identified limitations and enhance the practical applicability of the proposed model in real-world power systems. Additionally, incorporating automated reinforcement learning (RL) techniques [57] for hyperparameter optimization could further improve model performance and stability.

## CRediT authorship contribution statement

**Mehdi Taghizadeh:** Writing – original draft, Visualization, Validation, Methodology, Formal analysis, Data curation, Conceptualization. **Kamiar Khayambashi:** Writing – review & editing, Methodology, Data curation. **Md Abul Hasnat:** Writing – review & editing, Validation. **Negin Alemazkoor:** Writing – review & editing, Validation, Supervision, Resources, Project administration, Methodology, Funding acquisition, Conceptualization.

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

## Data availability

The code and data for this study are available in our GitHub repository at https://github.com/MehdiTaghizadehUVa/MFGNN_Power.

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